

Put I_3 in eqⁿ ①

$$I_1 + 3\left(\frac{7I_1 + I_2 - 1}{5}\right) - 5I_2 = 0$$

$$I_1 + \left(\frac{21I_1 + 3I_2 - 3}{5}\right) - 5I_2 = 0$$

$$5I_1 + 21I_1 + 3I_2 - 3 - 25I_2 = 0$$

$$26I_1 - 22I_2 = 3$$

$$13I_1 - 11I_2 = 1.5 \text{ --- ④}$$

Put I_3 in eqⁿ ②

$$5I_1 - I_2 - 9\left(\frac{7I_1 + I_2 - 1}{5}\right) = 0$$

$$25I_1 - 5I_2 - 63I_1 - 9I_2 + 9 = 0$$

$$-38I_1 - 14I_2 + 9 = 0$$

$$38I_1 + 14I_2 = 9$$

$$19I_1 + 7I_2 = \frac{9}{2} \text{ --- ⑤}$$

for eqⁿ ④

$$13I_1 - \frac{3}{2} = 11I_2$$

$$\frac{26I_1 - 3}{2} = 11I_2$$

$$26I_1 - 3 = 22I_2$$

$$\frac{26I_1 - 3}{22} = I_2$$

Put I_2 in eqⁿ (5)

$$19I_1 + 7\left(\frac{26I_1 - 3}{22}\right) = \frac{9}{2}$$

$$\frac{418I_1 + 182I_1 - 21}{22} = \frac{9}{2}$$

$$600I_1 - 21 = 99$$

$$600I_1 = 99 + 21$$

$$600I_1 = 120$$

$$I_1 = \frac{12}{600}$$

$$\boxed{I_1 = \frac{1}{5} \text{ A}}$$

For eqⁿ (4)

$$18I_1 - 11I_2 = \frac{3}{2}$$

$$18\left(\frac{1}{5}\right) - 11I_2 = \frac{3}{2}$$

$$\frac{18}{5} - \frac{3}{2} = 11I_2$$

$$\frac{26 - 15}{10} = 11I_2$$

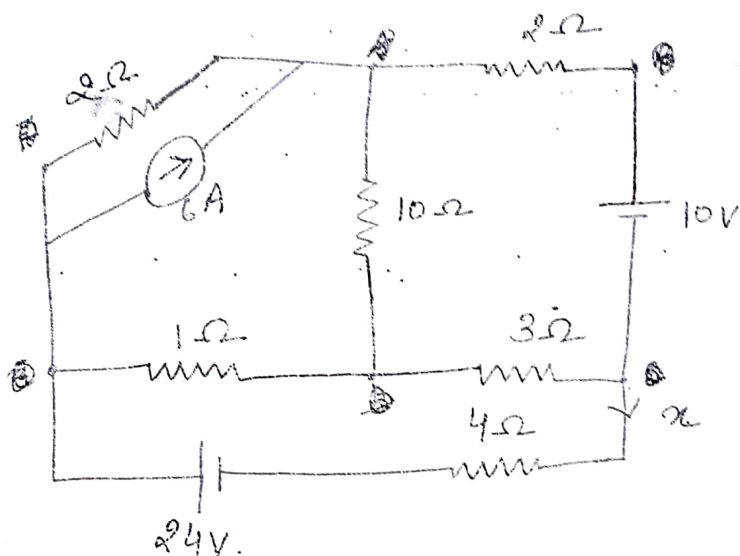
$$\frac{11}{10} = 11I_2$$

$$\boxed{I_2 = \frac{1}{10} \text{ A}}$$

$$I = I_1 + I_2 = \frac{1}{5} + \frac{1}{10} = \frac{2+1}{10} = \frac{3}{10} \text{ A}$$

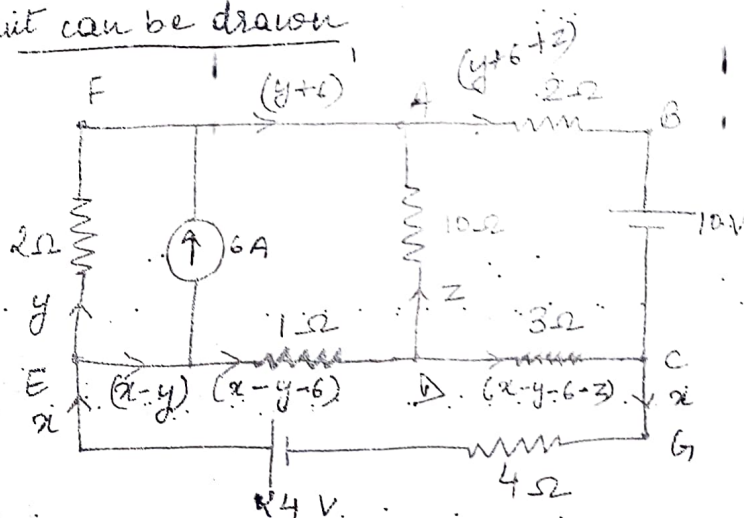
Q.13

Determine the current x in the 4Ω resistance in the circuit.



Solⁿ

Circuit can be drawn



In mesh EFAD E,

$$2y - 10z - (x - y - 6) = 0$$

$$2y - 10z - x + y + 6 = 0$$

$$3y - x - 10z + 6 = 0$$

$$x - 3y + 10z = 6 \quad \text{--- (1)}$$

In mesh ABCD A

$$2(y + z + 6) + 10 - 3(x - y - z - 6) + 10z = 0$$

$$2\check{y} + 2\check{z} + 12 + 10 - 3\check{x} + 3\check{y} + 3\check{z} + 18 + 10z = 0$$

$$-3x + 5y + 15z + 40 = 0$$

$$3x - 5y - 15z = 40 \quad (2)$$

Apply KVL in mesh EBCGE

$$(x - y - 6) + 3(x - y - z - 6) + 4x - 24 = 0$$

$$x - y - 6 + 3x - 3y - 3z - 18 + 4x - 24 = 0$$

$$8x - 4y - 3z - 48 = 0$$

$$8x - 4y - 3z = 48 \quad (3)$$

By matrix method

$$\begin{bmatrix} 1 & -3 & 10 \\ 3 & -5 & -15 \\ 8 & -4 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 40 \\ 48 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 1 & -3 & 10 \\ 3 & -5 & -15 \\ 8 & -4 & -3 \end{vmatrix} = 1(15 - 60) + 3(-9 + 120) + 10(-12)$$

$$= 45 + 3(-11) + 10(-28)$$

$$= 45 + 333 + 280$$

$$\Delta = 658$$

$$\Delta_1 = \begin{vmatrix} 6 & -3 & 10 \\ 40 & -5 & -15 \\ 48 & -4 & -3 \end{vmatrix} = 6(15 - 60) + 3(-120 + 720) + 10(-160 + 240)$$

$$= 6(45) + 3(600) + 10(80)$$

$$= 270 + 1800 + 800$$

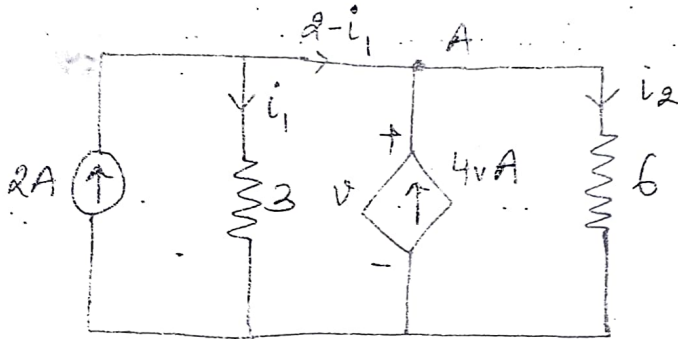
$$\Delta_1 = 2870$$

$$x = \frac{2870}{658} = 4.36 \text{ A}$$

$$\begin{array}{r} 180 \\ 120 \\ 270 \\ \hline 570 \end{array}$$

(using nodal analysis)

Q14. By applying Kirchhoff's current law, obtain the values of V , i_1 & i_2 in the circuit, which contains a voltage-dependent current source.



Soln

At node A

$$2 - i_1 + 4v = i_2$$

$$i_1 = \frac{V}{3}$$

$$i_2 = \frac{V}{6}$$

$$2 - \frac{V}{3} + 4v = \frac{V}{6}$$

$$\frac{V}{6} + \frac{V}{3} - 4v = 2$$

$$\frac{V + 2V - 24v}{6} = 2$$

$$3V - 24v = 12$$

$$-21v = 12$$

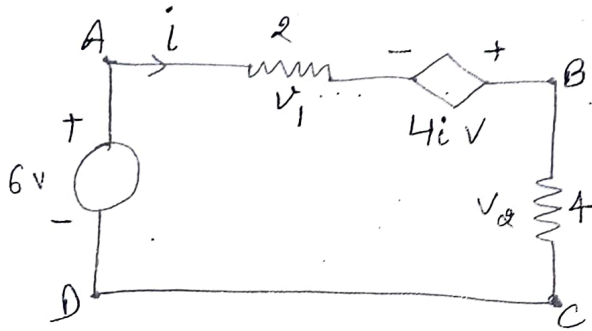
$$v = -\frac{12}{21} = -\frac{4}{7}$$

$$\boxed{V = -\frac{4}{7} \text{ V}}$$

$$i_1 = \frac{-\frac{4}{7}}{3} = -\frac{4}{21} \text{ A}$$

$$i_2 = \frac{V}{6} = \frac{-\frac{4}{7}}{6} = -\frac{4}{42} = -\frac{2}{21} \text{ A}$$

Q-15. Apply Kirchhoff's voltage law, to find the value of current i and the voltage drops V_1 & V_2 in the circuit which contains a current dependent voltage source. What is the voltage of dependent source?



Soln

Apply KVL

$$-6 + 2i - 4i + 4i = 0$$

$$2i = 6$$

$$i = 3A$$

$$V_1 = 2i = 6V$$

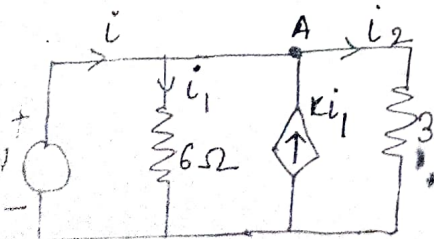
$$V_2 = 4i = 4 \times 3 = 12V$$

Voltage across dependent source $\rightarrow 4i = 12V$

Q-16. In the circuit, apply KCL to find the value of current i when (a) $k=2$

(b) $k=3$ &

(c) $k=4$



Soln

At node A

$$ki_1 + (i - i_1) = i_2$$

$$ki_1 + i - i_1 = i_2$$

$$(k-1)i_1 + i = i_2$$

$$(k-1)\frac{24}{6} + i = \frac{24}{3}$$

$$(k-1)4 + i = 8$$

$$4k - 4 + i = 8$$

$$i = 8 + 4 - 4k$$

(a) $k=2$

$$i = 12 - 8$$

$$i = 4A$$

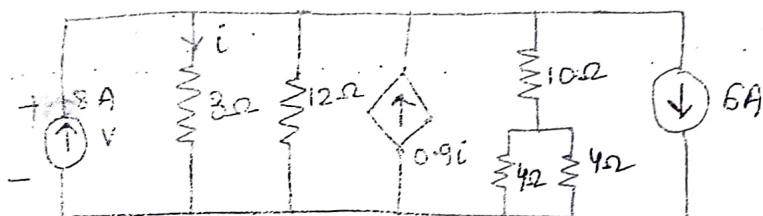
(b) $k=3$

$$i = 12 - 12$$

$$= 0$$

(c) $k=4$

Find the current i and also the power and of the dependent source in fig.



soln

The two current sources can be combined into a single source of $8-6 = 2A$.

Two parallel ($4\Omega || 4\Omega$) resistances gives

$$\frac{4 \cdot 4}{4+4} = \frac{16}{8} = 2\Omega$$

then in series with $10\Omega = 12\Omega$

then ($12\Omega || 12\Omega$)

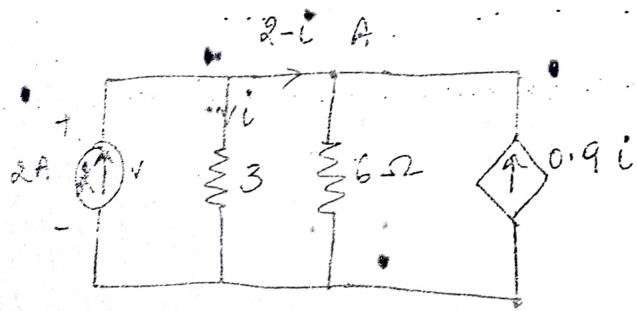
$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{R_1 + R_2}{R_1 R_2}$$

$$= \frac{1}{12} + \frac{1}{12} = \frac{12+12}{12 \cdot 12}$$

$$R = \frac{R_1 R_2}{R_1 + R_2}$$

$$\text{equivalent } R = \frac{12 \cdot 12}{12+12} = \frac{144}{24} = 6\Omega$$

Simplified ckt



At node A

$$2 - i + 0.9i = \frac{V}{6}$$

$$V = 3i$$

$$2 - i + 0.9i = \frac{3i}{6}$$

$$2 - 0.1i = \frac{3i}{6}$$

$$12 - 0.6i = 3i$$

$$12 = 3.6i$$

$$i = \frac{12}{3.6} = \frac{10}{3} A$$

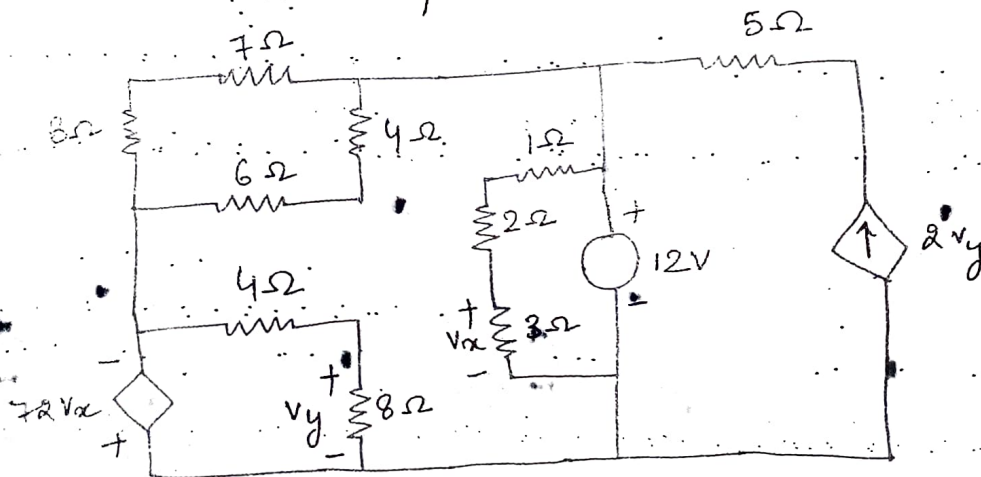
$$i = 10/3 \text{ A}$$

$$\begin{aligned} \text{voltage of dependent source} &= 10i \\ &= 10 \times \frac{10}{3} \text{ V} \end{aligned}$$

$$= \frac{3 \times 10}{3} = 10 \text{ V}$$

$$\begin{aligned} \text{Power of dependent source} &= V \times i \\ &= 10 \times 0.9i \\ &= 10 \times 0.9 \times \frac{10}{3} \\ &= 10 \times \frac{9}{10} \times \frac{10}{3} \\ &= 30 \text{ W} \end{aligned}$$

99
By voltage divider rule, calculate voltages V_x and V_y in the n/w



Soln

For V_x ,

$$V_x = \frac{3}{3+2+1} \times 12 = \frac{3}{6} \times 12 = 6 \text{ V}$$

$$72V_x = 72 \times 6 = 432 \text{ V}$$

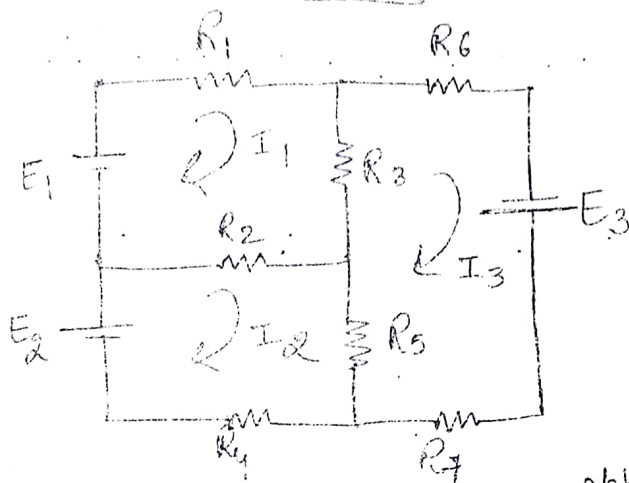
$$V_y = \frac{8}{8+4} \times 432 = \frac{8 \times 432}{12} = 288 \text{ V}$$

$$V_y = -288 \text{ V}$$

MESH ANALYSIS USING

RESISTANCE MATRIX METHOD

MATRIX FORM



Consider the n/w contains resistances and n voltage sources having n meshes.

Let three mesh currents be designated as I_1, I_2, I_3 & all three may be assumed to flow in clockwise direction obtaining symmetry in mesh eqn.

Apply

KVL to mesh (I)

$$-E_1 + I_1 R_1 + R_3(I_1 - I_3) + R_2(I_1 - I_2) = 0$$

$$I_1 R_1 + I_1 R_3 - I_3 R_3 + I_1 R_2 - I_2 R_2 = E_1$$

$$I_1 (R_1 + R_2 + R_3) - I_2 R_2 - I_3 R_3 = E_1 \quad \text{--- (1)}$$

KVL in mesh (II)

$$-E_2 + R_2(I_2 - I_1) + R_5(I_2 - I_3) + R_4 I_2 = 0$$

$$I_2 R_2 - I_1 R_2 + I_2 R_5 - I_3 R_5 + R_4 I_2 = E_2$$

$$I_2 (R_2 + R_5 + R_4) - I_1 R_2 - I_3 R_5 = E_2 \quad \text{--- (2)}$$

KVL in mesh (III)

$$I_3 R_7 + R_3(I_3 - I_2) + R_3(I_3 - I_1) + R_6 I_3 + E_3 = 0$$

$$I_3 R_7 + I_3 R_3 - I_2 R_3 + I_3 R_3 - I_1 R_3 + R_6 I_3 + E_3 = 0$$

$$I_3 (R_7 + R_3 + R_6) - I_2 R_3 - I_1 R_3 + E_3 = 0 \quad \text{--- (3)}$$

Equivalent matrix will be

$$\begin{bmatrix} R_1 + R_2 + R_3 & -R_2 & -R_3 \\ -R_2 & R_2 + R_4 + R_5 & -R_5 \\ -R_3 & -R_5 & R_3 + R_5 + R_6 + R_7 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} E_1 \\ E_2 \\ E_3 \end{bmatrix}$$

Sign of E_m : If we pass from -ve to +ve terminal of battery, then E_m is taken as +ve.

First Row

1st term: $(R_1 + R_2 + R_3)$ represents self resistance of mesh (i) which equals the sum of resistances in mesh (i).

2nd term: represents mutual resistance b/w mesh (i) & (ii). = -ve sum of the resistances common to mesh (i) & (ii).

3rd term: represents mutual resistance of mesh (i) & (iii).

E_1 : represents ~~sum~~ algebraic sum of voltage of all the voltage sources acting around mesh (i).

In general,

R_{11} = self resistance of mesh (i)

R_{22} = self resistance of mesh (ii)

R_{33} = " " (iii)

$R_{12} = R_{21} = -$ [sum of all the resistances common to mesh (i) & (ii)]

$R_{23} = R_{32} = -$ [sum of all the resistances common to mesh (ii) & (iii)]

$R_{31} = R_{13} = -$ [sum of all the resistances common to mesh (i) & (iii)]

Using these symbols, generalized form of matrix equivalent can be written as

$$\begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} E_1 \\ E_2 \\ E_3 \end{bmatrix}$$

Q.1. n no. of independent meshes;

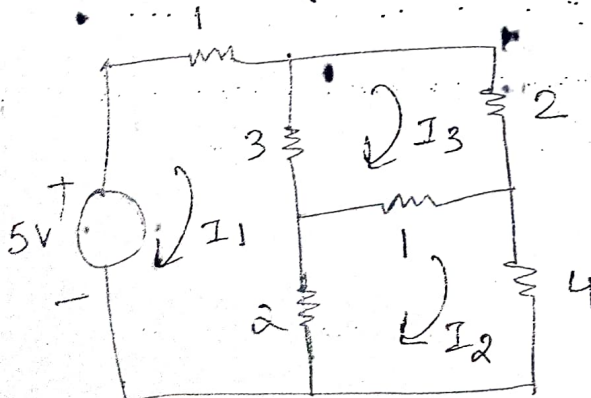
$$\begin{bmatrix} R_{11} & R_{12} & \dots & R_{1n} \\ R_{21} & R_{22} & \dots & R_{2n} \\ \dots & \dots & \dots & \dots \\ R_{n1} & R_{n2} & \dots & R_{nn} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ \dots \\ I_n \end{bmatrix} = \begin{bmatrix} E_1 \\ E_2 \\ \dots \\ E_n \end{bmatrix}$$

$$[R_m][I_m] = [E_m]$$

Ohm's law in matrix form

CSE-2
Q.1

Write ~~resistance~~ matrix of the n/w & find value of current I_3 .



mesh ①

$$R_{11} = 1 + 3 + 2 = 6 \Omega$$

$$R_{22} = 7 \Omega$$

$$R_{33} = 6 \Omega$$

$$R_{12} = R_{21} = -2 \Omega$$

$$R_{13} = R_{31} = -3 \Omega$$

$$R_{23} = R_{32} = -1 \Omega$$

$$E_1 = +5V$$

$$E_2 = 0$$

$$E_3 = 0$$

Matrix form

$$\begin{bmatrix} 6 & -2 & -3 \\ -2 & 7 & -1 \\ -3 & -1 & 6 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}$$

$$\Delta = \begin{bmatrix} 6 & -2 & -3 \\ -2 & 7 & -1 \\ -3 & -1 & 6 \end{bmatrix} = 147 = 0 \text{ (crossed out)}$$

$$\Delta_3 = \begin{bmatrix} 6 & -2 & 5 \\ -2 & 7 & 0 \\ -3 & -1 & 0 \end{bmatrix} = 115 = 0 \text{ (crossed out)}$$

$$I_3 = \frac{\Delta_3}{\Delta} = \frac{115}{147}$$

$I_3 = \frac{115}{147}$

$$\Delta_3 = 6(0) + 2(0) + 5(2+21) = 5(23) = \underline{\underline{115}}$$

MAXWELL'S LOOP CURRENT METHOD

MAXWELL'S CIRCULATING CURRENT THEOREM

→ This method which is particularly well suited to coupled circuit solutions employs a system of loop or mesh currents instead of branch currents (as in Kirchhoff's laws).

Here, the currents in different meshes are assigned continuous paths so that they do not split at junction into branch currents.

→ This method eliminates a great deal of tedious work involved in the branch-current method and is best suited when energy sources are distributed sources rather than a current source.

→ The method consists in writing a loop equation for each independent voltage law (KVL) loop in the network.

No. of independent equations = $b - (j - 1)$

where, b : no. of branches

j : no. of junctions

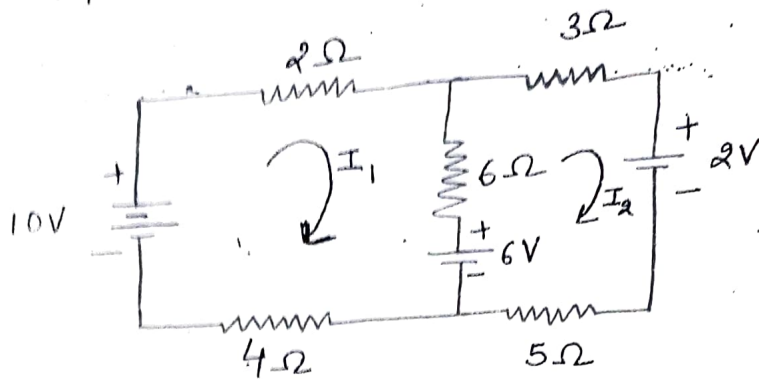
No. of independent equations = No. of meshes

→ In no. of cases, a network may be considered as consisting of a set of adjoining loops.

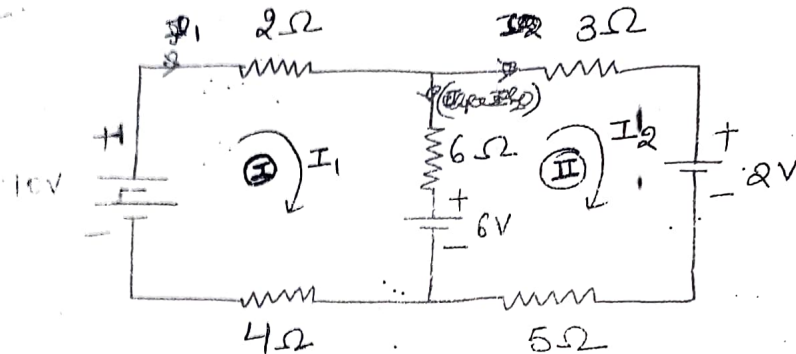
The theorem involves depicting entirely a current that is assumed to circulate around a closed loop by a curved arrow and labelling the arrow with its identifying current symbol I . By this theorem the current flowing through the branch common to two meshes will be equal to the algebraic sum of the two ~~mesh~~ currents flowing through it.

The dirⁿ of ~~mesh~~ current may be taken as clockwise or counter-clockwise but for systematic soln, dirⁿ of all ~~mesh~~ currents must be same.

Q.1 Using loop current method, find the current I_1 and I_2 as shown in fig.



SB12



Method 1

Loop I: Apply KVL

$$-10 + 2I_1 + 6(I_1 - I_2) + 6 + 4I_1 = 0$$

$$-10 + 2I_1 + 6I_1 - 6I_2 + 6 + 4I_1 = 0$$

$$12I_1 - 6I_2 - 4 = 0$$

$$6I_1 - 3I_2 - 2 = 0 \quad \text{--- (1)}$$

Loop II: Apply KVL

$$3I_2 + 2 + 5I_2 - 6 - 6(I_1 - I_2) = 0$$

$$3I_2 + 2 + 5I_2 - 6 - 6I_1 + 6I_2 = 0$$

$$14I_2 - 6I_1 - 4 = 0$$

$$7I_2 - 3I_1 - 2 = 0 \quad \text{--- (2)}$$

Method 2

for Loop I

Same procedure, I_1 dominates, so in middle branch current is $(I_1 - I_2)$.

for Loop II

current I_2 dominates and in middle branch current is $I_2 - I_1$, eqⁿ will be

$$3I_2 + 2 + 5I_2 - 6 + 6(I_2 - I_1) = 0$$

from eqⁿ ①

$$6I_1 - 3I_2 = 2$$

$$6I_1 - 2 = 3I_2$$

$$\frac{6I_1 - 2}{3} = I_2 \quad \text{--- ③}$$

Put I_2 in eqⁿ ②

$$7\left(\frac{6I_1 - 2}{3}\right) - 3I_1 - 2 = 0$$

$$\frac{42I_1 - 14 - 9I_1 - 6}{3} = 0$$

$$42I_1 - 9I_1 - 20 = 0$$

$$33I_1 = 20$$

$$I_1 = \frac{20}{33} \text{ A} = 0.606$$

Put I_1 in eqⁿ ③

$$\frac{26\left(\frac{20}{33}\right) - 2}{3} = I_2$$

$$\frac{(40 - 22)}{11} = I_2$$

$$I_2 = \frac{18}{23} \text{ A}$$

$$I_2 = \frac{6}{11} \text{ A} = 0.545$$

Using Resistance matrix method

$$R_{11} = 12 \Omega$$

$$R_{22} = 14 \Omega$$

$$R_{12} = R_{21} = -6 \Omega$$

$$E_1 = 10 - 6 = 4 \text{ V}$$

$$E_2 = 6 - 2 = 4 \text{ V}$$

$$\begin{bmatrix} 12 & -6 \\ -6 & 14 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

$$\Delta = \begin{bmatrix} 12 & -6 \\ -6 & 14 \end{bmatrix} = 12(14) - 36 = 168 - 36 = 132$$

$$I_1 = \frac{\begin{vmatrix} 4 & -6 \\ 4 & 14 \end{vmatrix}}{\Delta} = \frac{4(14) - 6(4)}{132} = \frac{56 - 24}{132} = \frac{32}{132} = 0.242$$

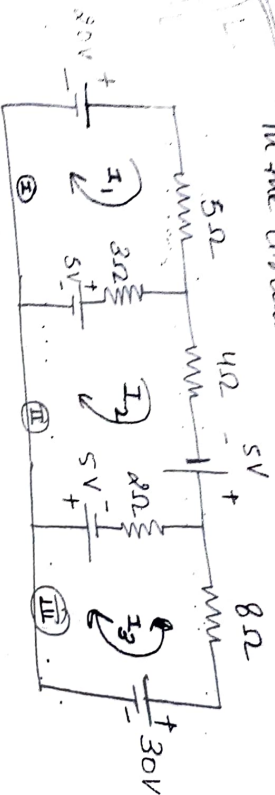
$$I_2 = \frac{\begin{vmatrix} 12 & 4 \\ -6 & 4 \end{vmatrix}}{\Delta} = \frac{12(4) - 4(-6)}{132} = \frac{48 + 24}{132} = \frac{72}{132} = 0.545$$

$$I_1 = \frac{32}{132} = 0.242 \text{ A}$$

$$I_2 = \frac{72}{132} = 0.545 \text{ A}$$

Q.2:

Determine the current supplied by each battery in the circuit.



Soln

For loop I:

$$5I_1 + 3(I_1 - I_2) + 5 - 20 = 0$$

$$5I_1 + 3I_1 - 3I_2 - 15 = 0$$

$$8I_1 - 3I_2 = 15 \quad \text{--- (1)}$$

For loop II:

$$4I_2 - 5 + 2(I_2 - I_3) - 5 - 5 + 3(I_2 - I_1) = 0$$

$$4I_2 + 2I_2 - 2I_3 + 3I_2 - 3I_1 - 15 = 0$$

$$9I_2 - 3I_1 + 2I_3 = 15 \quad \text{--- (2)}$$

For loop III:

$$+30 + 8I_3 + 2(I_3 - I_2) + 5 = 0$$

$$8I_3 - 2I_2 + 2I_3 + 35 = 0$$

$$-2I_2 + 10I_3 = -35 \quad \text{--- (3)}$$

By using Matrix Method

$$\begin{bmatrix} 8 & -3 & 0 \\ -3 & 9 & -2 \\ 0 & -2 & 10 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 15 \\ 15 \\ -35 \end{bmatrix}$$

For emf sign

! we pass from -ve to +ve

$$R_{11} = 8 \Omega$$

$$R_{22} = 9 \Omega$$

$$R_{33} = 10 \Omega$$

$$R_{12} = R_{21} = -3 \Omega$$

$$R_{13} = R_{31} = 0$$

$$R_{23} = R_{32} = -2 \Omega$$

$$E_1 = 20 - 5 = 15V$$

$$E_2 = 5 + 5 + 5 = 15V$$

$$E_3 = -5 - 30 = -35V$$

$$\Delta = \begin{bmatrix} 8 & -3 & 0 \\ -3 & 9 & -2 \\ 0 & -2 & 10 \end{bmatrix} = 8(90-4) + 3(-30) \\ = 8(86) - 90 \\ = 688 - 90$$

$$\boxed{\Delta = 598}$$

$$\Delta_1 = \begin{bmatrix} 15 & -3 & 0 \\ 15 & 9 & -2 \\ -35 & -2 & 10 \end{bmatrix} = 15(90-4) + 3(150-70) \\ = 15(86) + 3(80) \\ = 1290 + 240$$

$$\boxed{\Delta_1 = 1530}$$

$$\Delta_2 = \begin{bmatrix} 8 & 15 & 0 \\ -3 & 15 & -2 \\ 0 & -35 & 10 \end{bmatrix} = 8(150-70) - 15(-30) \\ = 8(80) + 450 \\ = 640 + 450$$

$$\boxed{\Delta_2 = 1090}$$

$$\Delta_3 = \begin{bmatrix} 8 & -3 & 15 \\ -3 & 9 & 15 \\ 0 & -2 & -35 \end{bmatrix} = 8(9 \times -35 + 30) + 3(105) + 5(6) \\ = 8(-315 + 30) + 315 + 90 \\ = 8(-285) + 405 \\ = -2280 + 405$$

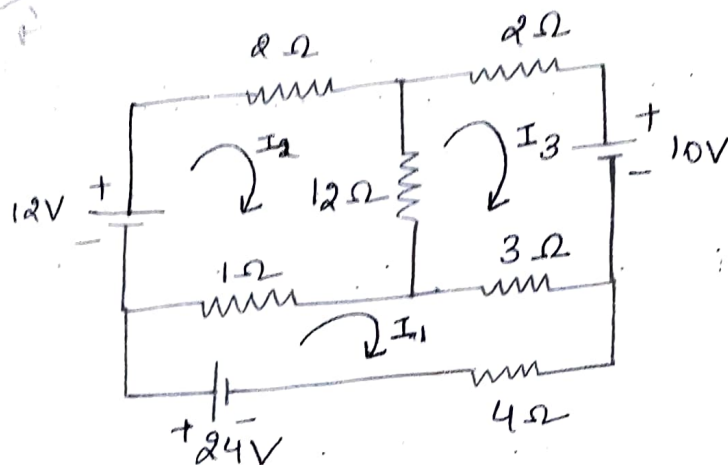
$$I_1 = \frac{1530}{598} = \frac{765}{299} \text{ A} = 2.55 \text{ A}$$

$$I_2 = \frac{1090}{598} = \frac{545}{299} \text{ A} = 1.822 \text{ A}$$

$$I_3 = \frac{-1875}{598} \text{ A} = -3.135 \text{ A}$$

$$\boxed{\Delta_3 = -1875}$$

Q-3 Determine the current in the 4Ω branch in the circuit.



Soln let current be in clockwise direction.

for loop I : Apply KVL

$$4I_1 - 24 + 1(I_1 - I_2) + 3(I_1 - I_3) = 0$$

$$4I_1 - 24 + I_1 - I_2 + 3I_1 - 3I_3 = 0$$

$$8I_1 - I_2 - 3I_3 = 24 \quad \text{--- (1)}$$

for loop II : Apply KVL

$$2I_2 + 12(I_2 - I_3) + 1(I_2 - I_1) - 12 = 0$$

$$2I_2 + 12I_2 - 12I_3 + I_2 - I_1 = 12$$

$$-I_1 + 15I_2 - 12I_3 = 12 \quad \text{--- (2)}$$

for loop III :

$$2I_3 + 10 + 3(I_3 - I_1) + 12(I_3 - I_2) = 0$$

$$2I_3 + 10 + 3I_3 - 3I_1 + 12I_3 - 12I_2 = 0$$

$$-3I_1 - 12I_2 + 17I_3 = -10 \quad \text{--- (3)}$$

From eqⁿ ①, ② & ③ - Using Matrix method.

$$\begin{bmatrix} 8 & -1 & -3 \\ -1 & 15 & -12 \\ -3 & -12 & 17 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 24 \\ 12 \\ -10 \end{bmatrix}$$

$$\Delta = \begin{bmatrix} 8 & -1 & -3 \\ -1 & 15 & -12 \\ -3 & -12 & 17 \end{bmatrix} = 8(15 \times 17 - 144) + 1(-17 - 36) - 3(12 + 45) \\ = 888 - 53 - 171 \\ = 888 - 224$$

$$\boxed{\Delta = 664}$$

$$\Delta_1 = \begin{bmatrix} 24 & -1 & -3 \\ 12 & 15 & -12 \\ -10 & -12 & 17 \end{bmatrix}$$

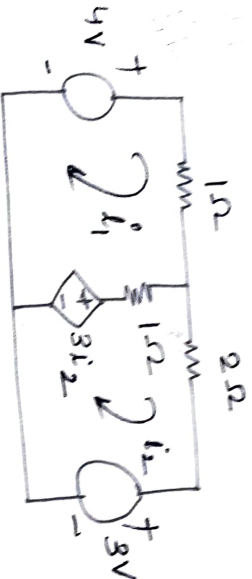
$$= 24(15 \times 17 - 144) + 1(12 \times 17 - 120) - 3(-144 + 150) \\ = 2664 + 84 - 18$$

$$\boxed{\Delta_1 = 2730}$$

$$I_1 = \frac{\Delta_1}{\Delta} = \frac{2730}{664} = 4.11 \text{ A}$$

$$\boxed{I_1 = 4.11 \text{ A}}$$

8-41: Find mesh currents i_1 and i_2 in the electric net.



Soln. KVL in mesh ①

$$-4 + i_1 + 1(i_1 - i_2) + 3i_2 = 0$$

$$i_1 + i_1 - i_2 + 3i_2 = 4$$

$$2i_1 + 2i_2 = 4$$

$$i_1 + i_2 = 2 \quad \text{--- ①}$$

KVL in mesh ②

$$2i_2 + 3 - 3i_2 + 1(i_2 - i_1) = 0$$

$$2i_2 + 3 - 3i_2 + i_2 - i_1 = 0$$

$$3 - i_1 = 0$$

$$i_1 = 3A$$

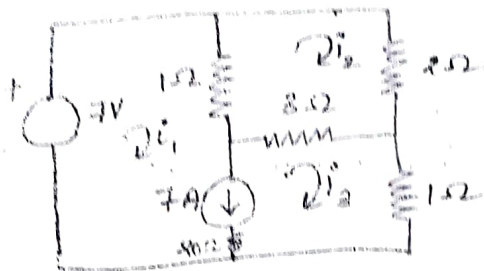
Put i_1 in eqn ①

$$3 + i_2 = 2$$

$$3 - 2 = -i_2$$

$$i_2 = -1A$$

SUPERMESH



Here, 7A is independent current source is in the boundary of 2 meshes & the current source comes to create the supermesh whose interior is that of meshes 1 & 3.

Whenever a current source is present at the boundary of two meshes so that we cannot apply the KVL eqn to individual meshes (like 1 & 3) i.e. we have to consider the loop that consist mesh 1 and mesh 3 for applying the KVL eqn. Such type of combination of meshes are known as SUPERMESH.

1st consider mesh ① & mesh ③ i.e. Supermesh & apply KVL

$$-7 + 1(i_1 - i_2) + 3(i_3 - i_2) + 1i_3 = 0$$

$$-7 + i_1 - i_2 + 3i_3 - 3i_2 + i_3 = 0$$

$$i_1 - 4i_2 + 4i_3 = 7 \quad \text{--- (1)}$$

Apply KVL to mesh ②

$$2i_2 + 3(i_2 - i_3) + 1(i_2 - i_1) = 0$$

$$2i_2 + 3i_2 - 3i_3 + i_2 - i_1 = 0$$

$$6i_2 - 3i_3 - i_1 = 0 \quad \text{--- (2)}$$

Value of 7A current source

$$7A = i_1 - i_3 \quad \text{--- (3)}$$

$$i_3 = i_1 - 7 \quad \text{--- (3)}$$

[Here, i_1 dominates because 7A current is in downward direction & dirn of i_1 is also in same way]



Put i_3 in eqⁿ ①

$$i_1 - 4i_2 + 4(i_1 - 7) = 7$$

$$i_1 - 4i_2 + 4i_1 - 28 = 7$$

$$5i_1 - 4i_2 = 35 \quad \text{--- (4)}$$

Put i_3 in eqⁿ ②

$$6i_2 - 3(i_1 - 7) - i_1 = 0$$

$$6i_2 - 3i_1 + 21 - i_1 = 0$$

$$-4i_1 + 6i_2 + 21 = 0$$

$$4i_1 - 6i_2 = 21 \quad \text{--- (5)}$$

$$4i_1 = 21 + 6i_2$$

$$i_1 = \frac{21 + 6i_2}{4}$$

Put i_1 in eqⁿ ④

$$5\left(\frac{21 + 6i_2}{4}\right) - 4i_2 = 35$$

$$105 + 30i_2 - 16i_2 = 140$$

$$105 + 14i_2 = 140$$

$$+14i_2 = 140 - 105$$

$$+14i_2 = 35$$

$$i_2 = 2.5 \text{ A}$$

Put i_2 in eqⁿ ⑤

$$4i_1 - 6(2.5) = 21$$

$$4i_1 - 15.0 = 21$$

$$4i_1 = 36$$

$$i_1 = 9 \text{ A}$$

Put i_1 in eqⁿ ③

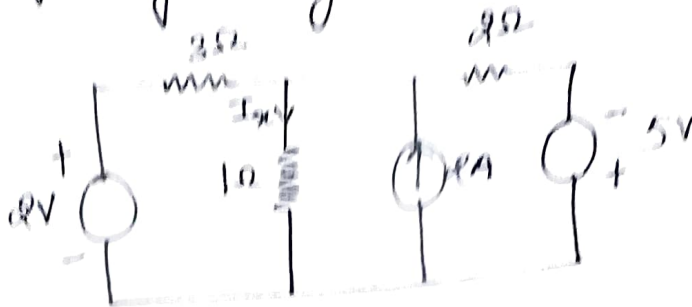
$$i_3 = i_1 - 7$$

$$= 9 - 7$$

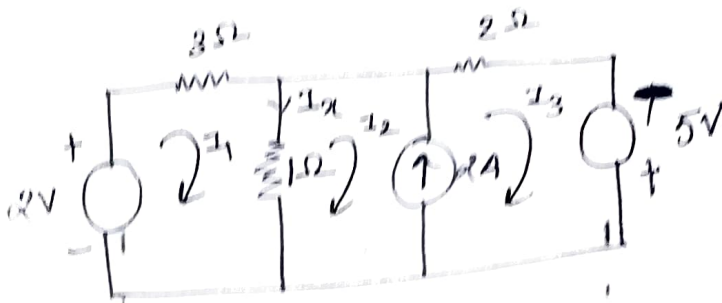
$$i_3 = 2 \text{ A}$$

Ques

Q2. Using mesh current method, determine current I_x in the following circuit.



Soln



KVL in mesh (1)

$$-2 + 3I_1 + 1(I_1 - I_2) = 0$$

$$4I_1 - I_2 = 2 \quad \text{--- (1)}$$

KVL in mesh (2) & mesh (3)

$$1(I_2 - I_1) + 2I_3 - 5 = 0$$

$$I_2 - I_1 + 2I_3 = 5 \quad \text{--- (2)}$$

$$2 = I_3 - I_2 \quad \text{--- (3)}$$

$$I_3 = 2 + I_2 \quad \text{--- (4)}$$

Put ' I_3 ' in eqn (2) & (3)

$$5 = I_2 + 2(2 + I_2) - I_1$$

$$5 = I_2 + 4 + 2I_2 - I_1$$

$$1 = 3I_2 - I_1 \quad \text{--- (5)}$$

Consider eqn (1) & (5)

$$4I_1 - I_2 = 2$$

$$I_2 = 4I_1 - 2$$

Put I_2 in eqn (5)

$$1 + I_1 = 3(4I_1 - 2)$$

$$1 = 12I_1 - 6 - I_1$$

$$1 = 11I_1 - 6$$

$$\boxed{I_1 = \frac{7}{11}}$$

Put ' I_1 ' in eqn (1)

$$4\left(\frac{7}{11}\right) - I_2 = 2$$

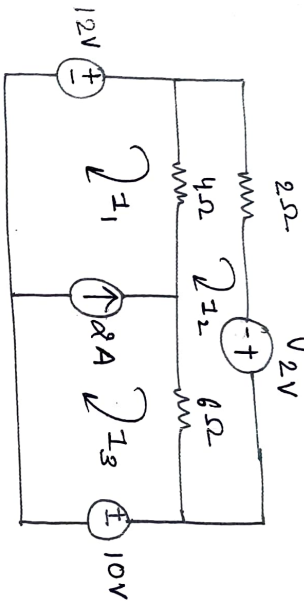
$$\frac{28}{11} - 2 = I_2$$

$$I_2 = \frac{28 - 22}{11} = \frac{6}{11}$$

$$I_x = I_1 - I_2 = \frac{7}{11} - \frac{6}{11} = \frac{1}{11}$$

$$\boxed{I_x = \frac{1}{11} \text{ A}}$$

Q-3- solve the ckt, using mesh method analysis & determine I_1, I_2 & I_3 . Evaluate power developed in the 10V voltage source.



Soln KVL in mesh ①

~~mesh 1~~ $2I_2 - 2 + 6(I_2 - I_3) + 4(I_2 - I_1) = 0$

$$8I_2 - 2 + 6I_2 - 6I_3 + 4I_2 - 4I_1 = 0$$

$$12I_2 - 6I_3 - 4I_1 = 2$$

$$6I_2 - 3I_3 - 2I_1 = 1 \quad \text{--- ①}$$

KVL in mesh ② & mesh ③

$$-12 + 4(I_1 - I_2) + 6(I_3 - I_2) + 10 = 0$$

$$-12 + 4I_1 - 4I_2 + 6I_3 - 6I_2 + 10 = 0$$

$$4I_1 - 10I_2 + 6I_3 = 2$$

$$2I_1 - 5I_2 + 3I_3 = 1 \quad \text{--- ②}$$

Current source

$$I_3 = I_3 - I_1 \quad \text{--- ③}$$

$$I_3 = 2 + I_1 \quad \text{--- ④}$$

$$6I_2 - 3I_3 - 2I_1 = 1 \quad \text{--- ①}$$

$$8I_1 - 5I_2 + 3I_3 = 1 \quad \text{--- ②}$$

$$2I_3 = 2 + I_1 \quad \text{--- ③}$$

Apply Cramer's rule.

$$\begin{bmatrix} -2 & 6 & -3 \\ 2 & -5 & 3 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$\Delta = \begin{bmatrix} -2 & 6 & -3 \\ 2 & -5 & 3 \\ -1 & 0 & 1 \end{bmatrix} = -2(-5) - 6(2+3) + (-3)(0-5) \\ = 10 - 6(5) - 3(-5) \\ = 10 - 30 + 15 \\ = 10 - 15 = -5$$

$$\boxed{\Delta = -5}$$

$$\Delta_1 = \begin{bmatrix} 1 & 6 & -3 \\ 1 & -5 & 3 \\ 2 & 0 & 1 \end{bmatrix} = 1(-5) - 6(1-6) - 3(0+10) \\ = -5 - 6(-5) - 3(10) \\ = -5 + 30 - 30$$

$$\boxed{\Delta_1 = -5}$$

$$\Delta_2 = \begin{bmatrix} -2 & 1 & -3 \\ 2 & 1 & 3 \\ -1 & 2 & 1 \end{bmatrix} = -2(1-6) - 1(2+3) - 3(4+1) \\ = -2(-5) - 1(5) - 3(5) \\ = 10 - 5 - 15 = 10 - 20 = -10$$

$$\boxed{\Delta_2 = -10}$$

$$\Delta_3 = \begin{bmatrix} -2 & 6 & 1 \\ 2 & -5 & 1 \\ -1 & 0 & 2 \end{bmatrix} = -2(-10) - 6(4+1) + 1(0-5) \\ = 20 - 30 - 5 \\ = 20 - 35 = -15$$

$$\boxed{\Delta_3 = -15}$$

$$I_1 = \frac{\Delta_1}{\Delta} = \frac{-5}{-5} = 1A$$

$$I_2 = \frac{\Delta_2}{\Delta} = \frac{-10}{-5} = 2A$$

$$I_3 = \frac{\Delta_3}{\Delta} = \frac{-15}{-5} = 3A$$

$$P_{(10V)} = V \times I \\ = 10 \times 3 \\ = 30W$$